Darwin Del Castillo, MD, MPH

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PROFESSIONAL SUMMARY

Epidemiologist and physician with an MPH in Epidemiology and extensive experience translating real-world evidence from claims, electronic health records, linked administrative data, cohorts, and observational studies across the UK, US, and Peru into meaningful insights. Strong background in causal inference, survival analysis, evidence synthesis, and communicating findings to both specialist and lay audiences. Adept at integrating state-of-the-art tools, such as large language models, into analytical workflows to support coding and research. Particularly interested in applying this skill set to advance non-communicable disease drug development in global, industry environments.

EDUCATION

Master of Public Health (MPH) – Epidemiology **Sep 2023 – Jun 2025**
University of Washington Seattle – USA

Doctor of Medicine (MD) **Mar 2014 – Dec 2020**
Universidad Nacional Mayor de San Marcos Lima – Peru

ADDITIONAL EDUCATION

Professional Certificate in Clinical Research **Sep 2021 – Apr 2022**
Universidad Peruana Cayetano Heredia Lima – Peru

SKILLS

Programming: R, Python, SQL (MySQL), SAS
Cloud Computing: Slurm HPC, Linux (bash)
Tools: Quarto, Shiny (R), REDCap, Git (GitHub, GitLab)
Data Standards: ICH E6, HIPAA, GDPR/UK GDPR, ICD-10/ICD-9, SNOMED CT
Packages: `tidyverse`, `survival`, `lme4`, `rstan`, `mice`, `cmprsk`, `ivreg`, `metafor`
Statistical methods: Bayesian hierarchical modelling (MCMC), GLM/GLMM, time-to-event models, marginal structural models, propensity score methods, competing risks, multiple imputation, meta-analysis, uncertainty estimation
Languages: English (fluent), Spanish (native)
Core Competencies: Real-world evidence generation, observational study design, scientific writing, stakeholder communication

PROFESSIONAL EXPERIENCE

Research Assistant – Epidemiology **Dec 2025 – Current**
School of Health and Wellbeing – University of Glasgow Glasgow – UK

- Estimating causal effects of social policy on health outcomes using survey data with g-methods.
- Modelling the direct and indirect effects of economic determinants of health on all-cause mortality using linked administrative data and doubly robust inference.
- Simulating policy scenarios using a dynamic stochastic model, delivering dashboards to be shared with key stakeholders.

Research Assistant – Clinical Informatics **Jul 2024 – Jun 2025**
Institute for Health Metrics and Evaluation Seattle – USA

- Developed inputs for a Bayesian model to estimate the burden and clinical outcomes of lower respiratory tract infections worldwide.
- Analysed large-scale RWD, including structured EHR data from 2 countries with 60+ million records, to generate epidemiological summaries of risk factors by sex.
- Developed MCMC-based pipelines with uncertainty propagation to estimate overlap among vulnerable population subgroups in 22 high-income countries to compute summary epidemiological measures.

RWD Analyst – Health Technology Assessment — **Jan 2023 – Jan 2024**
Social Security System (ESSALUD) — Lima – Peru

- Generated real-world evidence from harmonised claims (>5 million encounters) using ICD-10 phenotyping and probabilistic linkage.
- Built regression and survival analysis pipelines in R with quality control for reimbursement policy analyses and healthcare decision-making
- Developed analytical pipelines to evaluate country-level reimbursement policy options

Epidemiology Trainee — **Mar 2019 – Aug 2023**
CRONICAS Center of Excellence in Chronic Diseases — Lima – Peru

- Analysed longitudinal data from population cohorts and randomised trials in cardiometabolic epidemiology.
- Conducted analyses with biological biomarkers to support research grant proposals and scientific outputs.
- Co-authored 2 scientific manuscripts involving longitudinal analyses and machine learning models.

LICENSURE AND CERTIFICATION

National Registry of Investigators, RENACYT-CONCYTEC, Investigator Level VII (P0160977)	**2024**
CITI Program, Good Clinical Practice (ICH E6), Credential ID 65079238	**2023**
CITI Program, Human Subjects Learners, Credential ID 65079240	**2023**
CITI Program, Biomedical Responsible Conduct of Research, Credential ID 65079239	**2023**
Peruvian Medical Board, License No. 94534	**2021**

PUBLICATIONS ORCID: 0000-0002-8609-0312

Peer-Reviewed Publications

1. **Del Castillo-Fernández D**, Brañez-Condorena A, Villacorta-Landeo P, Saavedra-Garcia L, Bernabé-Ortiz A, Miranda J. Advances in the investigation of chronic non-communicable diseases in Peru. *An Fac med.* 2021; 81(4). DOI: 10.15381/anales.v81i4.18798
2. Lapidaire W, Proaño A, Blumenberg C, Loret de Mola C, Delgado CA, **Del Castillo D**, et al. Effect of preterm birth on growth and blood pressure in adulthood in the Pelotas 1993 cohort. *Int J Epidemiol.* 2023; 52(6):1870–7. DOI: 10.1093/ije/dyad084
3. Vidyasagaran AL, Ayesha R, Boehnke JR, Kirkham J, Rose L, Hurst JR, et al. Core outcome sets for trials of interventions to prevent and to treat multimorbidity in adults in low and middle-income countries: the COSMOS study. *BMJ Glob Health.* 2024; 9(8):e015120. DOI: 10.1136/bmjgh-2024-015120. Among authors: **Del Castillo Fernandez, D.**
4. Bazo-Alvarez JC, **Del Castillo D**, Piza L, Bernabé-Ortiz A, Carrillo-Larco RM, Smeeth L, et al. Multimorbidity patterns, sociodemographic characteristics, and mortality: Data science insights from low-resource settings. *Am J Epidemiol.* 2026; 194(4):901–11. DOI: 10.1093/aje/kwae466
5. Pinzon Cortes JA, Narongkiatikhun P, Fuhri Snethlage CM, **Del Castillo D**, Tommerdahl KL, Nadeau KJ, et al. Beyond glycemia in T1D: The hidden burden of insulin resistance and cardiorenal risk in type 1 diabetes - Part 1. *Endocrinol Metab Clin North Am.* 2026. DOI: 10.1016/j.ecl.2026.04.008
6. Pinzon Cortes JA, Narongkiatikhun P, Snethlage CF, **Del Castillo D**, Tommerdahl KL, Nadeau KJ, et al. Beyond glycemia in T1D: The hidden burden of insulin resistance and cardiorenal risk in type 1 diabetes - Part 2. *Endocrinol Metab Clin North Am.* 2026. DOI: 10.1016/j.ecl.2026.04.014